XINBO LUO

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EDUCATION

Rice University Houston, TX
Master of Electrical and Computer Engineering 08/2023 – 05/2025
GPA: 3.92/4.0
Relevant Courses: Deep Learning; Learning from Sensor Data; Advs in 3D Sensing & Vision

University of Liverpool Liverpool, UK
Bachelor of Engineering in Computer Science and Electronic Engineering 09/2021 – 07/2023
First Class Honours
Relevant Courses: Biocomputation; Neural Networks; Database Development; Signals and Systems

Xi'an Jiaotong-Liverpool University Suzhou, China
Bachelor of Engineering in Computer Science and Technology 08/2019 – 06/2021
First Class Honours
Relevant Courses: Linear Algebra; Multivariable Calculus; Engineering Mathematics

PUBLICATIONS

Liu, H., **Luo, X.**, Liu, H., & Xia, X. (2021). Merkle tree: A fundamental component of blockchains. In *2021 International Conference on Electronic Information Engineering and Computer Science (EIECS)*. IEEE. https://doi.org/10.1109/EIECS53707.2021.9588047

Research Interests: Statistical and machine learning methods for biomedical data; Biomedical signal analysis; human–LLM interaction; trust and evaluation of AI-assisted decision-making; AI for healthcare.

Programming: Python, R, SQL, MATLAB, C/C++, Java
Statistical and Machine Learning Methods: Linear regression; Decision trees; Random forests; Gradient boosting; Time-series modelling; Deep learning; Natural language processing;
AI Models and Agent Systems: LangChain, LangGraph, Hugging Face Transformers, LLM-based agents, retrieval-augmented generation (RAG), prompt engineering, tool calling, text embeddings, agent evaluation;
Libraries and Frameworks: TensorFlow, PyTorch, CatBoost, XGBoost, LightGBM, pandas, NumPy, scikit-learn;
Scientific Computing and Data Processing: Data pipelining; ETL workflows; Amazon Web Services (AWS); Amazon EC2; Google Cloud Platform (GCP).

TEACHING EXPERIENCE

Rice University Houston, TX
Teaching Assistant/George R. Brown School of Engineering 08/2023 – 05/2025

- Supported analog circuits laboratory instruction, guiding students through circuit construction, debugging, and verification of output waveforms;
- Assisted instruction in mobile and embedded systems, helping students with embedded Linux programming, device drivers, and hands-on implementation using Raspberry Pi hardware;
- Graded exams, laboratory reports, and assignments for both courses, assessing conceptual understanding and technical accuracy.

PROFESSIONAL EXPERIENCE

Sysco Corporation Houston, TX
Data Scientist 07/2025 – Present

- Integrated LLM agents with SQL and Python analytical tools to investigate margin erosion and generate evidence-backed pricing recommendations, reducing manual investigation;
- Developed an AI agent evaluation framework to measure tool-call accuracy, numerical correctness, and evidence grounding to identify failure modes and validate pricing recommendations;
- Built and deployed a pricing engine combining margin centroids, revenue-weighted modelling, product hierarchies, and customer purchasing behavior to generate pricing recommendations for the live rollout;
- Developed backend SQL queries powering production Tableau dashboards for the Canada Price Exception Monitoring project, which was nominated for the Sysco Technology Award;
- Designed an impact measurement framework for a rollout covering 42% of Canadian customers, comparing pre/post and test/control cohorts with stratified matching to quantify YoY GM$ growth;
- Refined pricing methodology through iterative cohort design, stratification and metric development; built an AI agent on the Sysco SAGE platform to transform pricing model outputs into executive-ready reports, informing Canada-wide pricing governance.

iClinic AI Inc. Houston, TX
Machine Learning Engineer 02/2025 – 06/2025

- Developed AI browsing agents by integrating vision-based tools with large language models (LLMs) to replicate human interactions with computer systems and execute natural language instructions – such as appointment scheduling – thereby reducing reliance on paid API integrations through interface-based automation;
- Created and designed dynamic conversation workflows of an inbound call answering agent on AI voice agent platforms by incorporating NLP techniques to structure coherent dialogues, resulting in improved patient engagement, streamlined clinical workflows, and increased operational efficiency;
- Collaborated with business and clinical stakeholders to translate market and user requirements into scalable technical solutions, effectively bridging engineering and healthcare needs.

Precision Insight Solutions Plano, TX
Data Scientist 01/2024 – 01/2025

- Worked as a DS consultant to help a restaurant client boost customer engagement by scraping posts and user data from the Rednote app, and identifying pain points and growth opportunities by analyzing social media trends;
- Designed and implemented a comprehensive, multi-faceted scoring system to evaluate potential Key Opinion Consumers (KOCs). Leveraged advanced NLP techniques – including TF-IDF, SimCSE, BERT for sentiment analysis, and ResNet-152 for domain classification – to assess originality, vertical domain relevance, sentiment intensity (with time-decay dynamics), and keyword coverage from scraped posts;
- Integrated diverse features such as followers age groups, posting frequencies, and sentiment trends into a dynamic scoring algorithm, enabling precise identification and ranking of high-impact influencers to optimize the restaurant's advertising strategy.

Joblogic-X Corporation Remote
Data Scientist 01/2023 – 10/2023

- Identified high-value features and extracted new features based on date/time analysis. Created bar plots, scatter plots, box plots and heatmaps to reveal patterns and relationships in the data;
- Built weekly sales forecasting models using ARIMA, Prophet and LightGBM across 50 stores, achieving a 9.8% MAPE (a 10% reduction versus baseline) and informing inventory decisions with projected annual cost savings of $40 million;
- Trained and evaluated classification models – AutoGluon, CatBoost, and XGBoost – to predict fraudulent transactions. Achieved a ROC-AUC score of 0.9061, significantly enhancing a local bank's fraud detection capabilities;
- Presented findings on both projects using time-series plots, feature importance graphs, and domain-specific visualizations to guide decision-making for fraud prevention strategies and inventory management.

RESEARCH EXPERIENCE

Statistical Analysis of rESWT for Post-Stroke Upper-Limb Spasticity 08/2026 – Present
Supervised by Dr. Peihua Cao, Southern Medical University

- Conducted biostatistical analysis of a two-arm randomized controlled trial evaluating radial extracorporeal shock wave therapy (rESWT) for post-stroke upper-limb spasticity;
- Performed data validation, exploratory data analysis, and missing-data assessment of longitudinal clinical data collected across four visits to characterize baseline characteristics and outcome distributions;
- Developed reproducible Python and R workflows to estimate treatment effects and compare outcomes between active-treatment and sham-control groups.

Heart Sound Classification Using Deep Learning 01/2024 – 12/2024
Supervised by Dr. Joseph Young, Rice University, in collaboration with HealthSeers

- Prepared and labeled phonocardiogram (PCG) recordings, applied FFT-based and wavelet filtering to reduce noise, and transformed recordings into Mel spectrograms for heart sound classification;
- Developed a TensorFlow-based convolutional neural network incorporating batch normalization and data augmentation to address overfitting and support generalization;

- Trained the model using Adam optimizer, achieving a 92% validation accuracy and ROC AUC of 0.87.

A Simulator for Protocols for Stable construction of Networks 09/2022 – 06/2023

Supervised by Professor Paul Spirakis, University of Liverpool

- Developed an interactive Python simulator using PyQT5, NetworkX to implement and visualize distributed network-construction protocols;
- Modeled random pairwise interactions and node and edge state transitions, implemented a heuristic stopping criterion to detect apparent stabilization enabling inspection of network formation and convergence;
- Evaluated simulated network structures against theoretical expectations and compared average convergence steps across repeated trials and varying network sizes, examining protocol behavior and the limitations of simulation-based stability detection.

Merkle Trees and Blockchain Data Verification 01/2021 – 06/2021

Supervised by Professor Vipul Goyal

- Completed cybersecurity coursework and contributed to a collaborative research paper examining Merkle trees as a component of blockchain systems;
- Contributed to a systematic overview of Merkle tree principles, properties, advantages and applications, focusing on their role in efficient and secure data verification;
- Coauthored “Merkle Tree: A Fundamental Component of Blockchains”, published in the proceedings of the 2021 international Conference on Electronic Information Engineering and Computer Science (EIECS).